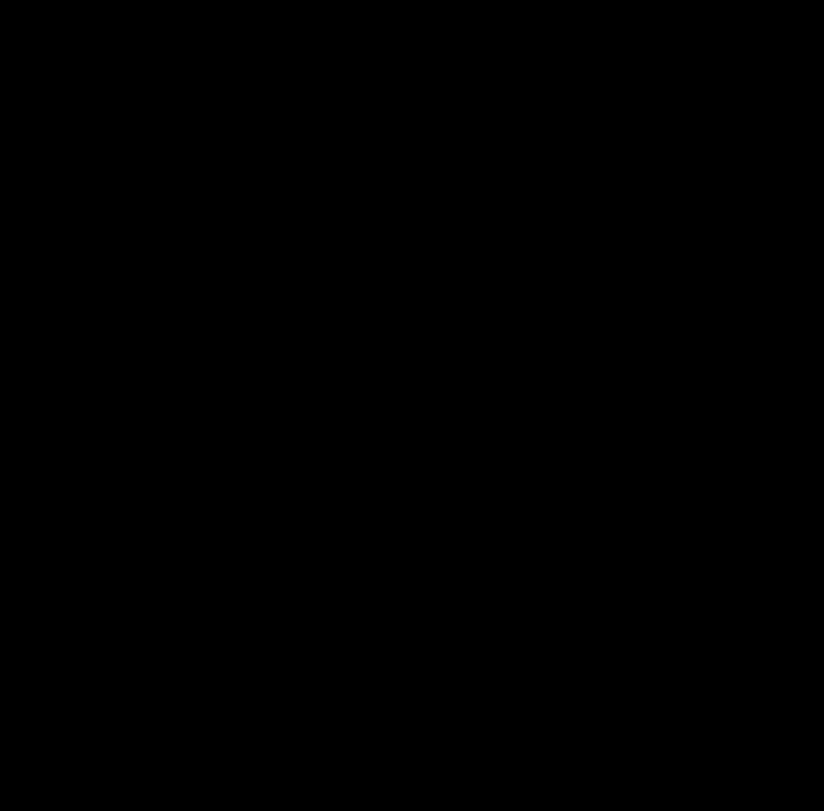


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INSTRUCTIONS TO CANDIDATES:

1. This Exam paper is composed of **three sections (A, B and C)**. Follow the instructions given below and answer the indicated questions for a total of **100 marks**.

Section **A**: Fourteen (14) questions which are **Compulsory** **55 marks**
Section **B**: Among the five (5) questions, attempt any three (3) **30 marks**
Section **C**: Among the two (2) questions, attempt anyone (1) **15 marks**
2. Fill in your actual **names** and **Index number** on the provided space (Cover).
3. Do not remove or tear any page or part of this Questions and Answers Booklet.
4. Answer in the language in which the Examination is set.
5. In section **A**, answer questions in the provided space. In case of multiple-choice questions, just **circle** the letter corresponding to the correct option.
6. In section **B** and **C**, answer questions in the provided space after section C questions.
7. **Allowed materials:**
 - Blue or Black pen.
 - Geometrical Instruments
 - Silent non-programmable calculators

SECTION A: Attempt all questions

(55 marks)

- 01.** A student is studying number systems and comes across the octal number system. The student claims that the digits used in the octal system are 0 through 7 and tries to use the digit 8 in a calculation. **(4marks)**

Determine Whether the Following Statements Are **True**(T) or **False**(F):

- a) The octal number system uses only the digits 0 to 7.....
- b) The digit 8 is a valid symbol in the octal number system.....

- 02.** A student is studying types of programming languages as part of his introductory computer science course. His teacher explains that programming languages can be categorized by how close they are to the computer's hardware. The student is given a worksheet to complete with the correct terms related to these types. **(4marks)**

Fill in the blanks using the correct terms:

- a) The programming language that consists of binary code and is directly executed by the CPU is called
- b) language uses symbolic code and is slightly more readable than machine language but still closely tied to hardware.
- c) Languages like Python and JavaScript that run using an interpreter at runtime are known as programming languages.
- d) C and C++ are examples of programming languages because they are translated into machine code by a compiler before execution.

- 03.** Ahmed is learning how to write and run a C program. His teacher explains that a complete program must be written with proper components and then go through a few essential steps before producing output. **(4marks)**

Determine Whether the Following Statements Are **True** or **False**:

- a) The final step in working with a C program is to debug it after successfully running it.....
- b) After writing the components of a C program, the next step is to compile the program to check for syntax errors.....

T2-146_Data Structure and Algorithms

- 04.** you are learning about the fundamental parts of a computer system and their functions. Your instructor gives you a set of fill-in-the-blank questions to test your understanding of basic computer hardware components. **(4marks)**

Fill in the blanks using the correct terms:

- a) The component of a computer that is often referred to as the "brain" because it executes instructions is the
- b) The component that stores data temporarily and is cleared when the computer is turned off is the.....
- c) The connects all hardware components and allows communication between them.
- d) The device that stores data permanently, even when the computer is off, is the.....

- 05.** As a student studying the concept of namespaces in C++ programming, your instructor explains how namespaces help manage the scope of identifiers such as functions, variables, and classes, especially in large programs or when integrating multiple libraries. **(4marks)**

Determine Whether the Following Statements are **True(T)** or **False(F)**:

- a) The primary role of namespaces in C++ is to group identifiers into different scopes, thereby helping to avoid name collisions.
- b) Namespaces in C++ are mainly used to define functions within classes, as indicated in the given material.

- 06.** You are reviewing core concepts in data structures and algorithms, specifically focusing on trees. Your instructor emphasizes the importance of recognizing the key properties that distinguish a tree from other types of graphs. You are asked to identify the correct characteristics of a tree. **(4marks)**

Which of the following statements, according to the provided material, are characteristics of a tree?

(Circle all that apply)

- a) It is a connected graph.
- b) It contains cycles.
- c) It is acyclic.
- d) It can be a forest.

T2-146_Data Structure and Algorithms

- 07.** You are working on designing a flowchart and writing an algorithm for a basic program. To complete your task correctly, you need to understand the purpose of various symbols and the rules that guide their use. Match each concept with its correct description or symbol. **(4marks)**

Match the items in Column A with the correct answers in Column B.

Column A (Concepts)	Column B (Descriptions)	Answer
1.Oval shape in a flowchart	a) Used to clearly define key steps and actions in the algorithm	1 =
2.Rectangular symbol in a flowchart	b) Shows the direction or flow of data or control	2 =
3.Arrow symbol in a flowchart	c) Represents an action or process step	3 =
4.Keywords like “Start”, “Input”, “Output” in an algorithm	d) Indicates the start or end of a process	4 =

- 08.** You are working on a C programming project that involves managing different data types using limited memory. Your instructor introduces the concept of unions and explains how they differ from structures. You are also shown examples of how enums can make your code more readable. You are now asked to demonstrate your understanding of these concepts. Which of the following statements correctly explain the behavior of unions, structures, and enums in C programming? **(5marks)**

(Circle all that apply):

- a) A union allows multiple members, but only one of them can hold a valid value at any one time.
- b) Accessing a union member that was not most recently written to may result in unexpected or corrupted values.
- c) Unlike structures, all union members share the same memory location.
- d) An enum in C is used to store a group of floating-point constants under symbolic names.
- e) A structure in C allows all its members to hold values independently and simultaneously.
- f) Union members can be accessed using the dot (.) operator when the union variable is declared normally (not as a pointer).

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- 09.** As an administrative assistant at Greenfield University, you have been assigned various tasks by different departments. Each task requires using specific features of text processing or tabular data processing software. **(4marks)**

Match each task with the correct feature or tool that should be used to complete it.

Task (Column A)	Features (Column B)	Answer
a) The English Department wants a well-formatted questionnaire that includes bullet point, tables for multiple-choice answers, and a university logo at the top.	1. Use Statistical Functions and Formulas in Excel or OpenOffice Calc	1 =
b) The Finance Office needs to calculate the average, median, and total expenditures of each department over the last quarter.	2. Use Image, List, and Table in MS Word or OpenOffice Writer.	2 =
c) The Dean's Office wants to present enrollment trends over five years using visual aids like bar chart and line graphs.	3. Create Hyperlinks Between worksheets in Tabular Software.	3 =
d) The IT Department wants a navigation sheet where clicking on department names opens specific data in different worksheets.	4. Use Charts and Graphs for Data Visualization in Tabular Software.	4 =

- 10.** As a software developer working on designing a software application for an e-commerce platform that must manage product listings, customer searches, and order processing, your goal is to choose data structures that suit the size of the data, how often it is accessed, and the operations required. **(4marks)**

Fill in the blanks below with the most appropriate factor influencing data structure choice. Choose your answers from the following options:

Options

- a) Performance Constraints
- b) Access Pattern
- c) Problem Requirements
- d) Data Size

statements

1. If the application must efficiently process millions of products, the most important factor to consider is
2. If customers frequently search for products using specific keywords, the influencing factor is the
3. The need to ensure orders are processed in the exact sequence they are received reflects a
4. Choosing between a binary tree or a hash table to ensure fast retrieval speed is driven by

- 11.** A Student is designing a flowchart to represent an algorithm that takes a user's age as input and then outputs whether the user is eligible to vote (age 18 or older) or not. Their initial flowchart draft looks like this: **(3marks)**

(Imagine a simple flowchart with the following elements, potentially drawn incorrectly or using non-standard symbols):

- A circle labeled "Start".
- A rectangle labeled "Get User Age".
- A diamond labeled "Age >= 18?".
- Two rectangles coming out of the diamond: one labeled "Print 'Eligible'" and the other labeled "Print 'Not Eligible'".
- A square labeled "End".
- Arrows connecting these shapes in a logical flow.

During a review session, you need to identify which aspects of their flowchart adhere to standard flowchart symbols and rules for this specific algorithm.

Which of the following statements correctly identify the appropriate use of symbols and rules in their flowchart draft? **(Circle all that apply)**

- a) The use of a circle labeled "Start" to indicate the beginning of the algorithm is a standard and appropriate symbol.
- b) The rectangle labeled "Get User Age" correctly represents an input operation in a flowchart.
- c) The diamond shape labeled "Age >= 18?" is appropriately used to represent a decision or conditional statement in the algorithm.
- d) The use of a square labeled "End" is the universally accepted symbol to denote the termination of a flowchart.
- e) Arrows should be used to connect all the symbols in the flowchart, indicating the flow of control and direction of the algorithm.

- 12.** A programming student is asked to write a function in C that calculates the product of two integers and returns the result. The student mistakenly writes the function as: **(4marks)**

```
void multiply (int a, int b) {  
    return a * b;  
}
```

The instructor asks the student to correct the mistakes and explain the correct version of the function.

Complete the statements below to correctly define and apply a C function that returns the product of two integers.

- a) To return a value from a function, the return type must not be, but should match the type of the value returned.
- b) The correct return type to use in this function is, because it returns an integer result.
- c) The function header should be written as:
..... multiply (int a, int b).
- d) The return statement in the function body should be:
..... a * b.

- 13.** A student is building a C++ program to track recently opened files in a text editor. The list should always show the most recently opened file first. To do this, the student uses a singly linked list and needs to insert each new file node at the beginning of list. **(3marks)**

Which of the following C++ function implementations correctly adds a new node with the given value at the beginning of a singly linked list?(circle the right option)

- a)

```
Node* addAtHead(Node* head, int value) {  
    Node* newNode = new Node(value);  
    newNode->next = head;  
    return newNode;  
}
```
- b)

```
Node* addAtHead(Node* head, int value) {  
    Node* newNode = new Node(value);  
    newNode->next = head->next;  
    head->next = newNode;  
    return head;  
}
```

T2-146_Data Structure and Algorithms

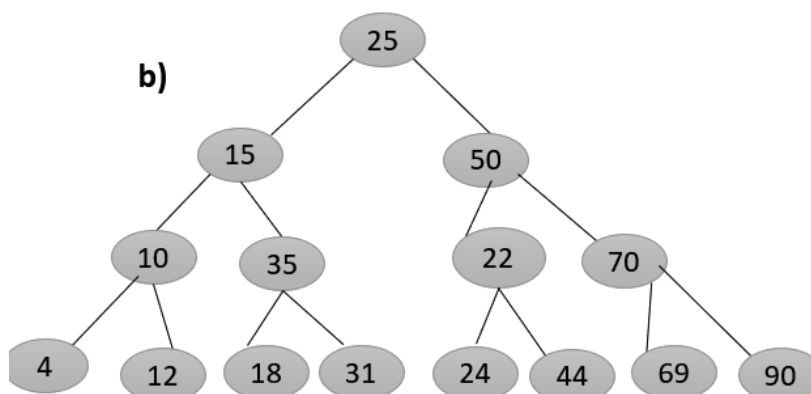
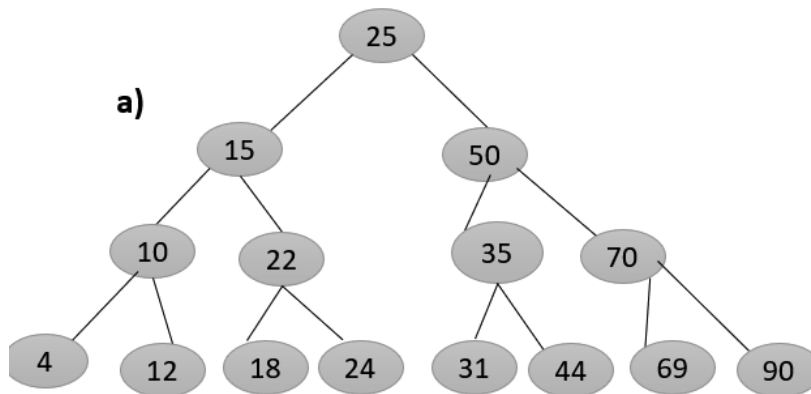
c) Node* addAtHead(Node* head, int value) {
Node* newNode = new Node(value);
newNode->next = head->next;
head->next = newNode;
return head;
}

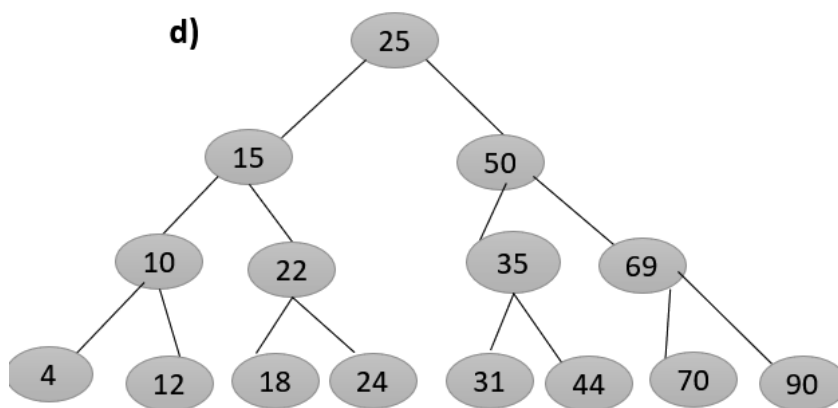
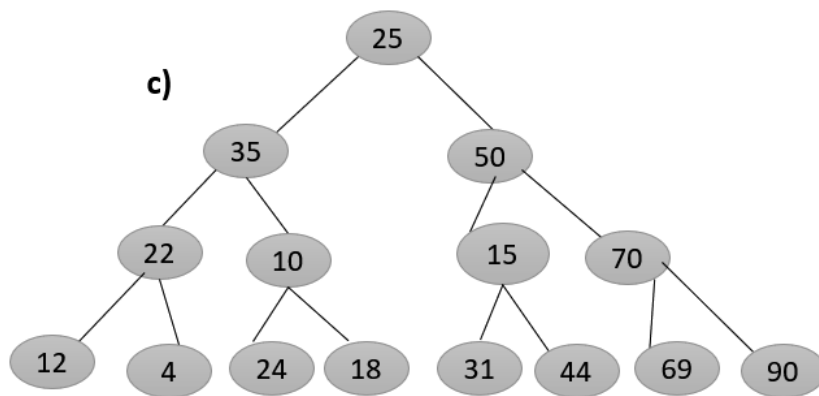
d) Node* addAtHead(Node* head, int value) {
Node* newNode = new Node(value);
head->next=newNode->nextl
newNode->next = head->next;
return head;
}

14. Circle the letter corresponding to the right answer to each of the following (4marks)
sub-questions related to this situation “You are given the following
numbers to insert into a Binary Search Tree (BST):

25, 50, 35, 15, 22, 31, 10, 70, 18, 12, 44, 4, 90, 24, 69”

- i) Which of the following represents the correct Binary Search Tree
formed after inserting the numbers in the given order?





ii) Which of the following correctly explains the three traversal methods of a Binary Tree?

- a)
- InOrder: Root → Left → Right
 - PreOrder: Left → Root → Right
 - PostOrder: Right → Left → Root
- b)
- InOrder: Right → Root → Left
 - PreOrder: Left → Right → Root
 - PostOrder: Root → Right → Left
- c)
- InOrder: Left → Root → Right
 - PreOrder: Root → Left → Right
 - PostOrder: Left → Right → Root
- d)
- InOrder: Left → Right → Root
 - PreOrder: Root → Right → Left
 - PostOrder: Left → Root → Right

Section B: Attempt any three (3) questions**(30 marks)**

- 15.** In a C program, a function is a block of code that performs a specific task. Functions are used to break down a program into smaller, manageable pieces. **(10marks)**

As learner, define a function in a C program which receives two numbers n_1 and n_2 as arguments and returns n_1 power n_2 .

N.B:

- The values of arguments are entered by the user at runtime
- Don't use the `pow()` function or any other pre-defined function which calculates the power of a number automatically.
- This is the example of program execution interface:

```
Enter two numbers:
5
3
5 power 3 is 125
```

- 16.** A university's data structures course is analyzing implementations of linked list operations. Students are examining various approaches to deleting the end node of singly linked when only the head pointer is provided. **(10marks)**

Task:

Write a function to delete the end of a Linked List when the head is provided. The signature is given below

`Node* deleteEnd(Node * head);`

- 17.** You are tasked with designing a recursive function to compute the sum of all elements in an integer array. Your function should handle arrays of any size, including the case where the array is empty. **(10marks)**

The function will take two parameters:

1. `array[]`: The array of integers.
2. `size`: The number of elements in the array.

Implement the function `recursiveSum()` to compute the sum of all elements in the array using recursion.

In your analysis, consider the following:

1. Base Case and Recursive Case: Identify how the problem should be broken down recursively, and define the base case for the recursion.

T2-146_ Data Structure and Algorithms

2. Handling Array Indexes: How will the array indices be managed in each recursive call?
3. Efficiency Considerations: Analyze the time complexity of your recursive solution.
4. Code Optimization: Consider whether the function can be optimized to prevent potential issues like stack overflow for large arrays

- 18.** You are working as a software engineer for an e-commerce platform that handles large amounts of customer data daily. The platform stores customer records, and each record includes the following information: **(10marks)**

- Customer ID
- Customer Name
- Account Creation Timestamp

The system needs to display the customers in chronological order based on their Account Creation Timestamp for generating reports and processing data. You have been tasked with selecting the best sorting algorithm for this task and implementing it.

The dataset consists of millions of records, and the sorting operation needs to be both efficient and stable. As part of the analysis, you are considering Merge Sort and comparing it against other sorting algorithms, such as Quick Sort and Heap Sort.

You need to evaluate Merge Sort based on the following criteria:

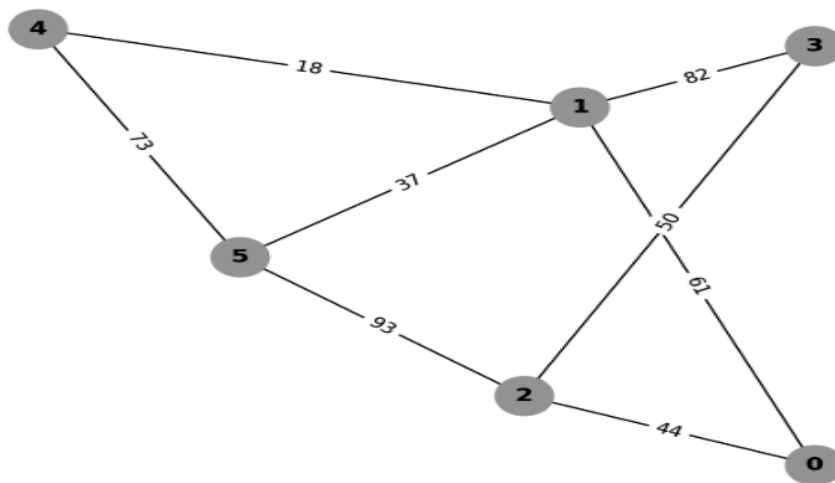
1. Time Complexity (worst-case, best-case, and average-case)
2. Space Complexity
3. Stability of the Algorithm
4. Handling Large Datasets (millions of records)
5. Practical Performance in terms of implementation ease and real-world usage

- 19.** A weighted graph G is a fundamental data structure in computer science where each edge connecting two vertices has an associated numerical value called a weight. These weights can represent various real-world quantities such as distances, costs, capacities, or priorities. **(10marks)**

Consider the weighted Graph G below and answer related questions.

- a) Apart from being weighted, indicates 5 types of graph that we can use to describe the Graph G .

b) Generate its adjacency matrix.



c) Generate the weighted degree of each node.

d) Write a c++ function that add an edge to a weighted graph.

Section C: Attempt only one (1) question**(15 marks)**

- 20.** As a junior developer at a software company that specializes in creating utility tools for internal use, your team leader has assigned you the task of developing a console-based tool that assists with digit-based data analysis. **(15marks)**

As a starting point, you are asked to create a C program which receives an integer number at run time and reverse it. For example, if the entered number is 523, the program will output 325.

- 21.** The dynamic queue implementation in C++ creates a FIFO structure using raw arrays that can grow automatically when needed. It requires manually coding essential operations (enqueue, dequeue, empty checks) and handling memory management. **(15marks)**

Question:

Provide an implementation of a dynamic queue operation using array in C++.

END

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